

State of Health Estimation Using a Novel Dataset Under Extreme Temperature and Aging Conditions

1st Yousef Alfoqaha

Department of Computer Engineering
German Jordanian University
y.mustafal@gnu.edu.jo

2nd Sahar Qadan

Department of Mechatronics Engineering
German Jordanian University
sahar.qadan@gnu.edu.jo

Abstract—Accurate State of Health (SoH) estimation in electric vehicle battery management systems is critical for operational safety. However, existing public benchmark datasets restrict model evaluation to narrow ambient temperatures and constant-current charging phases. To address this gap, a baseline SoH estimator was evaluated on a novel multi-protocol degradation dataset comprising eight Samsung INR21700-50E lithium-ion cells cycled across extreme temperatures (-20°C to 45°C) under realistic driving profiles, including the Worldwide Harmonised Light Vehicles Test Cycle, Hybrid Pulse Power Characterization, pulse, and constant-current segments. The proposed baseline framework combines 1D Convolutional Neural Networks, a Bidirectional Gated Recurrent Unit, and additive attention pooling, which achieved an overall SoH estimation error of 1.6% (RMSE = 0.016, $R^2 = 0.96$). Furthermore, unfreezing the attention pooling head during fine-tuning on the Oxford Battery Degradation Dataset demonstrates cross-dataset transferability, reducing fine-tuned error to 0.8% (RMSE = 0.008, $R^2 = 0.99$).

Index Terms—State of Health Estimation, Machine Learning, Lithium-Ion Batteries, Benchmark Dataset

I. INTRODUCTION

With the electric vehicle (EV) market share surpassing 20% of new global car sales [1], battery management systems (BMS) in EVs rely on accurate state of health (SoH) estimations to ensure a margin of safety in maintaining critical vehicle components. While SoH, in the capacity sense, is defined as $\text{SOH}_C = C/C_{\text{init}}$ [2], microscopic manufacturing variations cause each lithium-ion battery (LiB) to follow a highly non-linear degradation trajectory; this makes tracking SoH a non-trivial task, especially in realistic driving scenarios. Data-driven approaches typically employ diagnostic frameworks for real-time EV monitoring [3], [4]; by extracting instant features from available charge-discharge windows, they dynamically track sudden non-linear aging mechanisms like solid electrolyte interphase (SEI) layer growth and lithium plating as they occur [5].

Recent diagnostic models combine convolutional neural networks (CNN) for spatial feature extraction with recurrent neural networks (RNN), such as Long Short-Term Memory (LSTM) [6], [7] or Gated Recurrent Units (GRU) [8], [9], to model temporal dependencies common in LiB degradation. These networks frequently integrate attention mechanisms to weight cycle data [9], [10] and utilize support vector machines (SVM) or support vector regression (SVR) for final SoH regression [11], [12].

Because generating battery degradation data is a laborious and time-consuming process, the vast majority of these models are trained and benchmarked on open-datasets such as NASA Ames [13], Oxford [14], CALCE [15], TJU [16], and HUST [17]. While these datasets offer varied cell chemistries, they restrict model evaluation to constant-current (CC) charging phases and narrow ambient temperatures. Thus, to complement existing benchmark datasets, a multi-protocol LiB degradation dataset was developed in collaboration with the University of Wuppertal [18]. This dataset comprises 8 Samsung INR21700-50E LiBs [19], which were cycled across a wide temperature spectrum (-20°C to 45°C) using a combination of realistic operational protocols, including the Worldwide Harmonised Light Vehicles Test Cycle (WLTC) [20], Hybrid Pulse Power Characterization (HPPC) [21], pulse, and CC segments.

To establish a benchmark baseline, evaluation was performed on a diagnostic estimator combining CNN, BiGRU, and attention pooling layers. This baseline was evaluated across the dataset's temperature bands and protocols, and verify its cross-dataset transferability by fine-tuning it on the Oxford Battery Degradation Dataset. A comparative summary of public battery datasets and estimation frameworks is provided in Table I.

II. DATA PROCESSING

A. Dataset

The dataset was obtained from the Scienlab SL1007A charge-discharge testing platform, where eight Samsung INR21700-50E lithium-ion cells (Table II) were cycled from fresh SoH levels to beyond end-of-life [22]. Telemetry of each cell was recorded using Measurement and Control Units (MCU) and exported as ASAM Measurement Data Format (MF4) files [23].

TABLE II
SAMSUNG INR21700-50E CELL SPECIFICATION

Parameter	Value
Chemistry	NMC (Li-ion)
Form Factor	21700 Cylindrical
Nominal Capacity	4,900 mAh
Rated Capacity (C_{init})	5.0 Ah
Nominal Voltage	3.6 V
Voltage Range	2.5–4.2 V
Max. Charge Rate	2C
Max. Discharge Rate	4C

TABLE I
COMPARISON OF BENCHMARK BATTERY DEGRADATION DATASETS AND DIAGNOSTIC ESTIMATION FRAMEWORKS

Dataset / Study	Cell Form / Chemistry	Temperature Range	Discharge Protocols	Discharge Rate
NASA Ames	18650 Cylindrical LCO-NCA	20°C to 30°C	CC	1C
Oxford Dataset 1	Pouch / LCO (740 mAh)	40°C (Static)	CC	1C
HUST Dataset	18650 Cylindrical LFP	Uncontrolled Ambient	Multi-step CC	0.25–4C
This Work	21700 Cylindrical NMC (5.0 Ah)	−20°C to 45°C	WLTC, HPPC, Pulse, CC	0.1, 0.5, 1, 2C

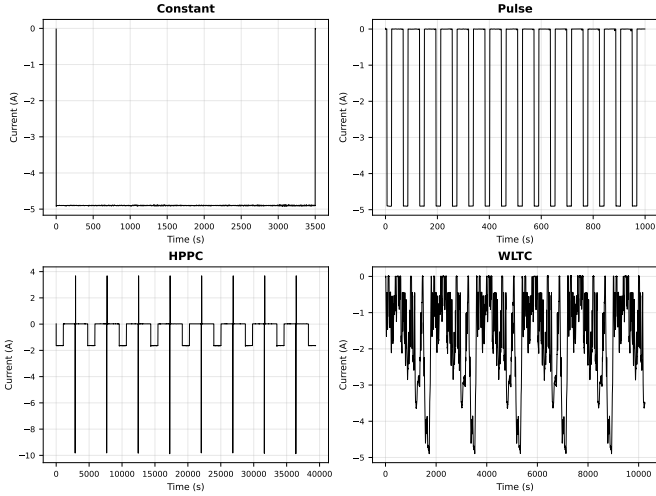


Fig. 1. Discharge profiles for the four test protocols: CC, Pulse, HPPC, and WLTC.

Initial measurements were taken across CC, pulse, HPPC, and WLTC protocols (Fig. 1), with ambient temperatures at -20 , -10 , -5 , 0 , 10 , 25 , and 40°C . CC and pulse protocols were tested at 0.1 , 0.5 , 1.0 , and 2.0C . Cyclic aging was then performed using CC discharge at 1.0C and 25°C . Post aging measurements were taken using the same procedure as the initial phase (excluding WLTC). The distribution of cycles across temperature bands and discharge protocols is shown in Table III.

TABLE III
TEMPERATURE AND PROTOCOL DISTRIBUTION

Temp Band	Constant	HPPC	Pulse	WLTC	Total
-20°C	15	16	46	0	77
-10°C	23	16	41	0	80
-5°C	0	6	0	0	6
0°C	25	22	81	0	128
10°C	0	22	0	0	22
25°C	914	22	64	14	1014
40°C	27	22	66	0	115
Total	1004	126	298	14	1442

B. Pre-processing Pipeline

As per the dataset engineer’s advice, voltage (V), current (I), temperature (T), and ambient temperature (T_{amb}) are primary

channels recorded from the test platform’s sensors. A script was used to filter discharge periods using start/end signals provided by the testing platform. Raw cycles recorded at up to 1 kHz (1 ms) were downsampled to 1 Hz (1 s). For each cycle, the mean value of T_{amb} was taken since it is a fixed condition set by the test platform.

The MF4 cycles were then consolidated into a single Hierarchical Data Format (HDF5) [24] file using h5py [25], structured as one HDF5 group per MCU containing a sub-group per discharge cycle. Each cycle sub-group stores V , I , and T as N -dimensional datasets. Metadata, including the discharge protocol, testing phase, ambient temperature, discharge rate, date-time, cycle index, and target SoH, is embedded directly as scalar attributes.

Consolidating the raw files into a single HDF5 container eliminates MF4 file-handling overhead, provides $O(1)$ random-access indexing for data loading, and presents a uniform interface for model training. This consolidation also reduced total storage from approximately 240 GB to 450 MB , as the raw MF4 files retained numerous derived channels and signals beyond the four primary channels (V , I , T , T_{amb}) used for training.

C. SoH Labeling

Raw SoH for each discharge cycle was computed via Coulomb counting:

$$\text{SoH}_{\text{raw}} = \frac{\int_{t_{\text{start}}}^{t_{\text{end}}} |I(t)| dt}{C_{\text{init}}} \quad (1)$$

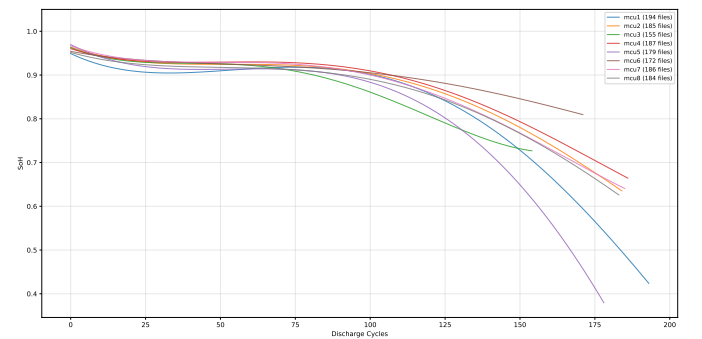


Fig. 2. Fitted SoH trajectories for all eight MCUs across their full discharge cycle histories.

Measured capacity depends on discharge rate and T_{amb} , thus raw SoH is not directly comparable across cycles. Discharge cycles at 1C and 25°C, the only condition common to all phases, were used as reference points. A 4th degree polynomial was fit over cycle index to these reference points, giving a per-MCU SoH trajectory (Fig. 2). Reference-point SoH ranges and cycle counts are given in Table IV.

TABLE IV
MCU SOH RANGE BY PHASE

MCU	Initial	Aging	Post-Aging	Cycles
1	94.3–93.4	92.5–67.4	66.7–47.9	194
2	95.6–94.1	93.0–78.2	78.6–67.8	185
3	97.4–94.6	95.6–80.9	82.5–74.0	155
4	95.3–95.0	93.4–79.4	79.8–71.1	187
5	95.7–94.3	92.1–66.2	65.0–42.3	179
6	95.8–93.3	93.4–86.9	87.6–82.7	172
7	95.8–95.0	93.3–76.6	78.5–67.5	186
8	94.6–94.1	92.3–76.9	76.4–66.8	184

D. Feature Scaling

Z-score standardization is sensitive to extreme temperature and aging conditions, where wide swings are not considered outliers. Instead, an unclipped robust scaling approach [26] was used to map the 1st and 99th percentiles (p_{01} , p_{99}) to $[-1, 1]$, normalizing the 14,262,367 time steps for training stability. Mean and standard deviation per feature are given in Table V.

$$x_{\text{norm}} = \frac{x - p_{01}}{p_{99} - p_{01}} \quad (2)$$

TABLE V
FEATURE STATISTICS

Feature	Mean (μ)	Std. Dev. (σ)
Voltage (V)	3.5997	0.3253
Current (I)	-1.5349	2.1722
Cell Temperature (T)	16.9000	18.0783
Ambient Temperature (T_{amb})	15.4799	18.0773
Thermal Delta (ΔT_{cell})	1.4201	1.7980
State of Health (SoH)	0.8579	0.1028

III. METHODOLOGY

A. Problem Formulation

Given a sequence of V , I , and T over a discharge cycle, conditioned on T_{amb} , an estimator function $f(\cdot)$ predicts the cell's SoH:

$$\widehat{\text{SoH}} = f(V, I, T \mid T_{\text{amb}}) \quad (3)$$

B. Estimator Architecture

As illustrated in Fig. 3, A window of sequences V , I , and T of length N were passed through a stem 1D-CNN layer followed by two strided 1D-CNN blocks (GroupNorm, LeakyReLU, dropout), followed by Feature-wise Linear Modulation (FiLM) [27] conditioning on T_{amb} . Since T_{amb} is fixed within a cycle while V , I , and T vary at every timestep, FiLM

was chosen to rescale and shift the CNN feature maps directly rather than treating ambient temperature as an additional sequential channel:

$$\text{FiLM}(h) = (1 + \gamma(T_{\text{amb}})) \odot h + \beta(T_{\text{amb}}) \quad (4)$$

where γ and β are learned affine parameters. The modulated features pass through a 2-layer Bidirectional GRU (BiGRU). An additive attention mechanism then pools over the BiGRU output timesteps:

$$\alpha_t = \frac{\exp(w^\top \tanh(Wh_t))}{\sum_\tau \exp(w^\top \tanh(Wh_\tau))}, \quad z = \sum_t \alpha_t h_t \quad (5)$$

where h_t is the BiGRU output at timestep t . The pooled representation z was projected to a scalar SoH estimate.

C. Training Setup

Raw input sequences (1,500 to 30,000 timesteps per cycle) were sliced into 1,500-timestep windows (15 minutes) with a 1,000-timestep stride. The model predicted the SoH for each window, and these predictions were averaged to produce a single SoH value per cycle. The eight MCUs were split 6/1/1 into training, validation, and test sets. MCUs 2 and 7 were held out for validation and testing, respectively, as their SoH range (95.6–67.5%) falls within the training range (97.4–42.3%). The validation set was shuffled once to break temporal symmetry between the initial, aging, and post-aging phases. The training set was shuffled every epoch.

In total, the training set comprised 1,071 cycles (7,227 windows), while the validation set contained 185 cycles (1,341 windows). All input sequences and conditions were normalized using robust min-max scaling. The Estimator's hyperparameters are given in Table VI.

TABLE VI
ESTIMATOR HYPERPARAMETERS

Hyperparameter	Value
Conv. base channels	64
Conv. kernel size	7
Conv. stride	2
GRU hidden size	128
GRU layers	2
Dropout	0.2
LeakyReLU slope	0.2

The framework was implemented in PyTorch and accelerated on an NVIDIA RTX 5060 Ti 16GB. The model was trained using the AdamW optimizer (weight decay of 10^{-2}) and a batch size of 128 windows to minimize the mean squared error (MSE) loss. The ReduceLROnPlateau scheduler was used on an initial learning rate of 2×10^{-2} , which decayed by a factor of 0.5 (minimum 10^{-5}) upon encountering a 15-epoch plateau in validation loss. Early stopping was used to halt training after 50 epochs without improvement, restoring the model weights that achieved the lowest validation loss.

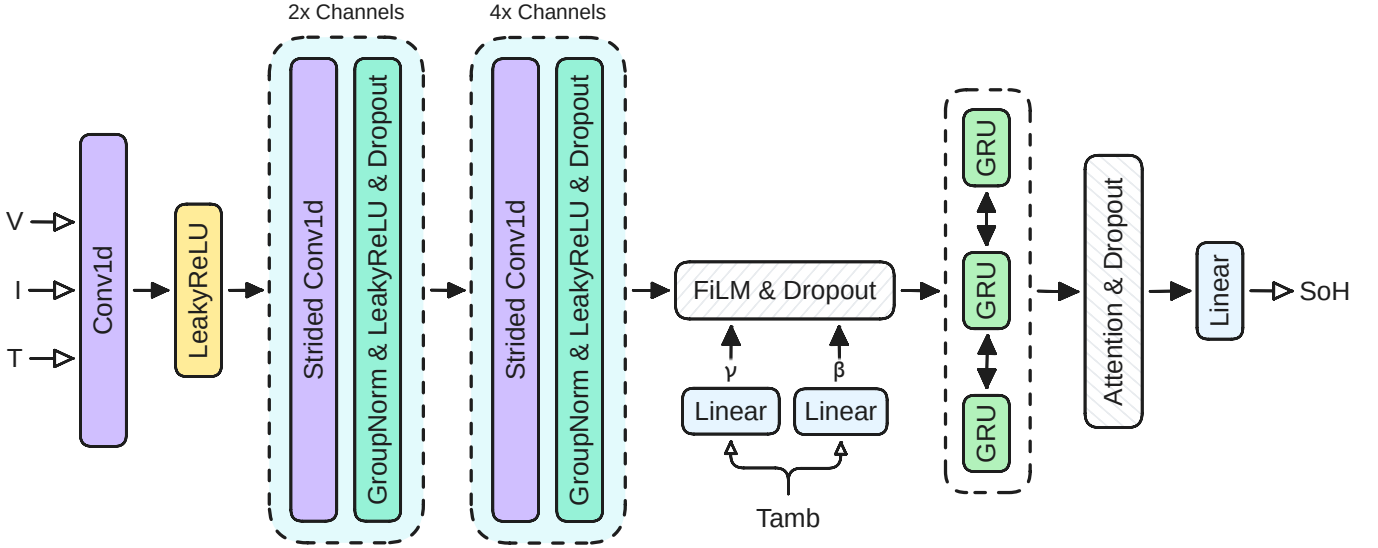


Fig. 3. Model architecture: 1D-CNN blocks, FiLM conditioning on T_{amb} , BiGRU, and attention pooling.

D. Results

Evaluation was performed on the testing MCU across testing phases, temperature bands, and discharge protocols (Table VII). The initial phase reached 1.3% error (RMSE 0.015, $R^2 = -0.81$). Aging was the strongest phase ($R^2 = 0.94$, 1.1% error, RMSE 0.011). Post-Aging error rose to 3.0% (RMSE 0.031). Overall, $R^2 = 0.96$ and 1.6% error were achieved across 62.8–96.2% SoH.

The 40°C band achieved a high fit among temperature slices ($R^2 = 0.98$, 1.5% error), while -20°C trailed behind with the lowest fit ($R^2 = 0.83$, 4.0% error) and -10°C showed a strong fit among the higher-cycle bands ($R^2 = 0.98$, 2.4% error). -20°C spanned 68.1–93.7% at $R^2 = 0.83$ and 40°C spanned 66.8–95.1% at $R^2 = 0.98$. The 0°C and 25°C bands, the two largest by cycle count (18 and 129, respectively), achieved $R^2 = 0.98$ and 0.96 . The -5°C and 10°C bands were represented by only 1 and 3 cycles, respectively.

Across discharge protocols, each protocol spanned multiple temperature bands (Table III). Constant and HPPC achieved the highest fits ($R^2 = 0.97$ and 0.99), and Pulse followed ($R^2 = 0.93$, 1.9% error). WLTC reported an R^2 of 0.10 (2 cycles, both restricted to the Initial phase) despite a low 1.3% error.

The predicted trajectory in Fig. 4 closely tracked the fitted SoH curve through the initial and aging phases, with predictions staying within a narrow band of the reference trajectory up to approximately cycle 125. Prediction variance increased visibly in the Post-Aging region, consistent with the elevated post-aging RMSE in Table VII, with individual predictions deviating further from the fitted trajectory beyond approximately cycle 150.

TABLE VII
BASELINE ESTIMATOR PERFORMANCE ACROSS ALL SLICES

Slice	RMSE	MAE	R^2	%Err	Cycles
Testing Phases					
Initial	0.015	0.012	-0.81	1.3%	48
Aging	0.011	0.010	0.94	1.1%	99
Post-Aging	0.031	0.021	0.43	3.0%	39
Temperature Bands					
-20°C	0.050	0.029	0.83	4.0%	9
-10°C	0.018	0.017	0.98	2.4%	11
-5°C	0.026	0.026	–	2.8%	1
0°C	0.017	0.014	0.98	1.8%	18
10°C	0.010	0.008	0.99	0.9%	3
25°C	0.014	0.011	0.96	1.3%	129
40°C	0.016	0.012	0.98	1.5%	15
Discharge Protocols					
Constant	0.015	0.013	0.97	1.5%	125
HPPC	0.013	0.010	0.99	1.2%	17
Pulse	0.026	0.015	0.93	1.9%	42
WLTC	0.012	0.012	0.10	1.3%	2
Overall	0.018	0.013	0.96	1.6%	186

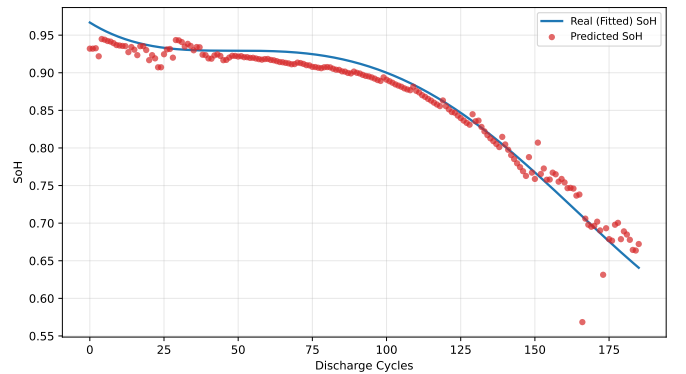


Fig. 4. Testing MCU SoH trajectory with predicted SoH points.

IV. CROSS-DATASET VALIDATION

A. Oxford Battery Degradation Dataset 1

The model was fine-tuned on the Oxford Battery Degradation Dataset 1 to evaluate cross-dataset transferability. The dataset comprises eight Kokam 740 mAh LCO pouch cells (SLPB526495) aged under constant 1C discharge at 40°C. This presents a severe domain shift in cell architecture, chemistry, and capacity relative to the Samsung INR21700-50E cylindrical cells (5.0 Ah) used for primary training. SoH labels were computed via Coulomb counting (1). SoH ranges for each cell are given in Table VIII.

TABLE VIII
OXFORD CELL SOH RANGE AND CYCLE COUNT

Cell	SoH Range (%)	Cycles
1	99.9–75.7	78
2	99.3–67.8	73
3	99.3–76.5	76
4	99.2–78.5	47
5	99.4–61.6	46
6	99.0–78.2	46
7	98.5–78.8	77
8	98.3–75.4	76

B. Fine-Tuning Setup

The 1D-CNN, FiLM, and BiGRU layers were frozen, leaving only the attention pooling mechanism and output regression head open for fine-tuning. The eight cells were partitioned into a 6/1/1 split, allocating Cells 1–6 for fine-tuning, Cell 7 for validation, and Cell 8 for testing (Table VIII). Feature scaling bounds, hyperparameter settings and shuffling strategy were kept identical to the primary training setup.

C. Results

Evaluating the model in a zero-shot setting yielded severe performance degradation, resulting in an RMSE of 0.406 (46.8% error) and an R^2 of -36.66 . However, unfreezing only the attention pooling mechanism and output regression head resolved this domain discrepancy. Upon fine-tuning, performance improved dramatically, achieving an RMSE of 0.0080, an MAE of 0.0060, an R^2 of 0.99, and reducing the mean percentage error to 0.8% across all 76 evaluation cycles. As shown in Fig. 5, in contrast to the zero-shot predictions, the fine-tuned predictions closely tracked ground-truth capacity fade.

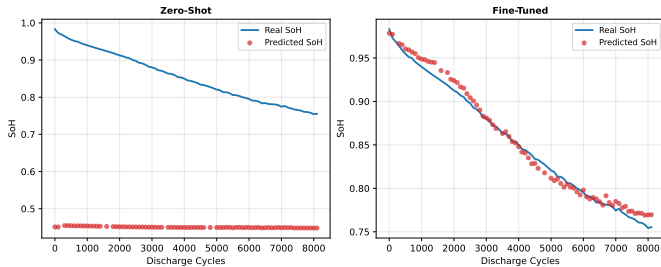


Fig. 5. Estimated versus ground-truth SoH trajectory on the testing cell for both zero-shot and fine-tuned scenarios.

V. FEATURE INTERPRETABILITY

A. Permutation Feature Importance (PFI)

PFI was used to measure which inputs the model relies on for its predictions. PFI shuffles one feature at a time across discharge cycles and measures the resulting drop in performance:

$$\Delta\text{RMSE}_j = \text{RMSE}_{\text{perm}(j)} - \text{RMSE}_{\text{base}}, \quad j \in \{I, V, T, T_{\text{amb}}\} \quad (6)$$

Shuffling was only done within the following controlled testing conditions: discharge protocol, ambient temperature, and discharge rate. This isolates the true importance of each feature from differences caused by the test setup itself.

B. Results

The stratified PFI results (Fig. 6) demonstrate that current (I) and voltage (V) are the primary drivers of model predictions across all protocols.

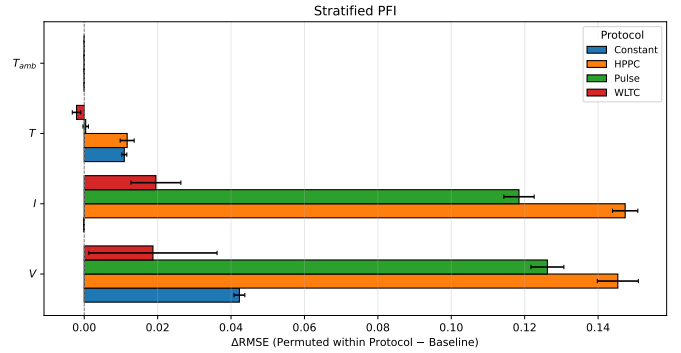


Fig. 6. Protocol-stratified Permutation Feature Importance (ΔRMSE) across input channels (I, V, T, T_{amb}).

VI. DISCUSSION

The model performed best on the Aging phase, likely due to its narrow test condition of constant 1C discharge at 25°C. The Initial and Post-Aging phases instead demanded the model evaluate SoH across a wide temperature range and all discharge protocols. Possible reasons for the temperature-driven error include the raw deviation from the well-represented 25°C condition, and slower ion kinetics making SoH harder to distinguish at extremes. RMSE was comparable across all protocols despite training data being dominated by CC discharge, suggesting the model learned degradation signatures that survive.

The Oxford fine-tuning results revealed that zero-shot transfer failed ($\text{RMSE} = 0.406$) due to the severe capacity and chemistry mismatch between 5.0 Ah NMC cylindrical and 740 mAh LCO pouch cells. However, fine-tuning only the attention pooling and output head resolved this almost entirely ($\text{RMSE} = 0.008$, $R^2 = 0.99$). This suggests the backbone learned universal temporal degradation features, while the attention head re-calibrated to the new capacity scale.

Recently, Li et al. [6] conducted a comprehensive comparison of several data-driven architectures on the Oxford Degradation

Dataset 1, evaluating their ACLA hybrid model against established Neural-ODE (NODE and ANODE) baselines.

TABLE IX
STATE-OF-HEALTH ESTIMATION PERFORMANCE COMPARISON ON OXFORD
DATASET 1

Model / Framework	Study	RMSE
ANODE	Li et al. [6]	0.0117
NODE	Li et al. [6]	0.0113
ACLA	Li et al. [6]	0.0093
CNN-BiGRU-Attn	This Work	0.0080

As detailed in Table IX, the fine-tuned model outperformed the cited models, achieving a lower RMSE of 0.0080. However, the models evaluated by Li et al. rely on manually extracting and discretizing normalized time features from isolated CC charging phases, whereas this work processed raw timestep windows (V, I, T) directly. This may suggest that models can interpret degradation indicators present in raw telemetry.

The PFI results reinforced this: V and I dominated across all protocols, aligning with SEI-driven internal resistance growth. Thermal reliance shifted dynamically, rising under CC discharge ($\Delta\text{RMSE} = +0.0130$) where sustained current caused measurable heat accumulation, and dropping under HPPC and Pulse where rest periods allow localized dissipation. This suggests the model tracked physical degradation signatures rather than dataset-specific artifacts.

VII. CONCLUSION

This work introduced a multi-protocol LiB degradation dataset spanning extreme temperatures (-20° to 45°C) and realistic discharge profiles, addressing the narrow conditions of existing benchmarks. A CNN-BiGRU-Attention baseline achieved 1.6% SoH error ($R^2 = 0.96$) overall, and 0.8% ($R^2 = 0.99$) after fine-tuning on the Oxford dataset, outperforming existing Neural-ODE baselines despite a severe domain shift. PFI confirmed current and voltage as the dominant predictive features, indicating the model learned physically grounded, transferable degradation signatures.

Future work includes generative augmentation of underrepresented operating conditions, ablation studies against simpler RNN baselines, and finer-grained interpretability via SHAP.

REFERENCES

- [1] International Energy Agency, "Global EV Outlook 2024: Trends in electric car markets," <https://www.iea.org/reports/global-ev-outlook-2024>, 2024.
- [2] C. R. Birkel, M. R. Roberts, E. McTurk, P. G. Bruce, and D. A. Howey, "Degradation diagnostics and prognosis of lithium-ion batteries," *Journal of Power Sources*, vol. 341, pp. 373–386, 2017.
- [3] X. Hu, Y. Zhang, X. Tang, X. Lin, R. Zhang, and S.-Y. Choe, "Advanced fault diagnosis technologies for a lithium-ion battery system: A review," *Renewable and Sustainable Energy Reviews*, vol. 121, p. 109689, 2020.
- [4] M. S. H. Lipu, M. A. Hannan, A. Hussain, M. S. H. Hoque, P. J. Ker, and Z. Y. Dong, "A review on data-driven state of health estimation for lithium-ion batteries in electric vehicles: Challenges and future trends," *IEEE Transactions on Industry Applications*, vol. 59, no. 2, pp. 2202–2214, 2023.
- [5] S. E. J. O’Kane, W. Ai, G. Madabattula, D. A. Alonso, R. Timms, V. Sulzer, J. S. Edge, B. Wu, and G. J. Offer, "Lithium-ion battery degradation: how to model it," *Physical Chemistry Chemical Physics*, vol. 24, no. 13, pp. 7909–7922, 2022.
- [6] Y. Li, M. He, and J. Liu, "A novel Neural-ODE model for the state of health estimation of lithium-ion battery using charging curve," *arXiv preprint arXiv:2505.05803*, 2025.
- [7] D. Chen, X. Zheng, C. Chen, and W. Zhao, "Remaining useful life prediction of the lithium-ion battery based on CNN-LSTM fusion model and grey relational analysis," *Mathematical Biosciences and Engineering*, vol. 31, no. 1, pp. 633–655, 2023.
- [8] F. Xia, K. Wang, and J. Chen, "State of health and remaining useful life prediction of lithium-ion batteries based on a disturbance-free incremental capacity and differential voltage analysis method," *Journal of Energy Storage*, vol. 64, p. 107161, 2023.
- [9] X. Chen, Y. Qin, W. Zhao, Q. Yang, N. Cai, and K. Wu, "A self-attention knowledge domain adaptation network for commercial lithium-ion batteries state-of-health estimation under shallow cycles," *arXiv preprint arXiv:2304.05084*, 2023.
- [10] Y. Wei and D. Wu, "Prediction of state of health and remaining useful life of lithium-ion battery using graph convolutional network with dual attention mechanisms," *Reliability Engineering & System Safety*, vol. 230, p. 108947, 2023.
- [11] X. Zhang, J. Feng, F. Cai, K. Huang, and S. Wang, "A novel state of health estimation model for lithium-ion batteries incorporating signal processing and optimized machine learning methods," *Frontiers in Energy*, vol. 19, no. 3, pp. 348–364, 2025.
- [12] R. Li, N. He, and F. Cheng, "State of health estimation of lithium-ion battery based on novel health indicators and improved support vector regression," *Batteries*, vol. 11, no. 10, p. 347, 2025.
- [13] B. Saha and K. Goebel, "Battery Data Set," NASA Ames Prognostics Data Repository, NASA Ames Research Center, Moffett Field, CA, 2007. [Online]. Available: <https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/>
- [14] C. R. Birkel, "Oxford Battery Degradation Dataset 1," University of Oxford, Bodleian Data Repository, 2017.
- [15] Center for Advanced Life Cycle Engineering (CALCE), "CALCE Battery Research Group Datasets," <https://calce.umd.edu/battery-data>, 2011.
- [16] F. Wang, Z. Zhai, Z. Zhao, X. Chen, and R. Yan, "Physics-informed neural network for lithium-ion battery degradation stable modeling and prognosis," *Nature Communications*, vol. 15, no. 1, p. 4332, 2024.
- [17] W. Li, W. Sha, S. Tang, S. Lu, and Y.-C. Cao, "A large-scale lithium-ion battery dataset for capacity estimation and remaining useful life prediction," Mendeley Data, V2, 2021.
- [18] University of Wuppertal, "Institute of Electric Mobility and Energy Storage Systems (EES)," <https://ees.uni-wuppertal.de/en/>, 2026, bergische Universität Wuppertal, Germany.
- [19] Samsung SDI Co., Ltd., *Specification of Product for Lithium-ion Rechargeable Cell Model: INR21700-50E*, revision 1.0 ed., Samsung SDI Co., Ltd., 2018, technical Datasheet.
- [20] United Nations Economic Commission for Europe, *Global Technical Regulation No. 15: Worldwide Harmonized Light Vehicles Test Procedures (WLTP)*, UN ECE UN Global Technical Regulation ECE/TRANS/180/Add.15, 2014. [Online]. Available: <https://unece.org/GTR-15>
- [21] Idaho National Engineering and Environmental Laboratory, *Battery Test Manual for Plug-In Hybrid Electric Vehicles*, U.S. Department of Energy, FreedomCAR & Vehicle Technologies Program Technical Report DOE/ID-11163, 2010. [Online]. Available: <https://avt.inl.gov/sites/default/files/pdf/phev/phevTestManualRev2.pdf>
- [22] Keysight Technologies, *SL1007A Scienlab Battery Test System – Cell Level*, Keysight Technologies, 2024, data Sheet 5992-4247EN. [Online]. Available: <https://www.keysight.com/us/en/product/SL1007A/sl1007a-sciencelab-battery-test-system-cell-level.html>
- [23] ASAM e.V., *ASAM MDF 4.1.0: Measurement Data Format*, Association for Standardisation of Automation and Measuring Systems Technical Standard, 2012. [Online]. Available: <https://www.asam.net/standards/detail/mdf/>
- [24] The HDF Group, *Hierarchical Data Format, Version 5*, The HDF Group Data Specification, 1997–2026. [Online]. Available: <https://www.hdfgroup.org/HDF5/>
- [25] A. Collette, "Python and HDF5: Fast Indexing and Storage with Data Structures," O’Reilly Media, 2013.
- [26] J. Han, M. Kamber, and J. Pei, *Data mining: concepts and techniques*, 3rd ed. Morgan Kaufmann, 2011.
- [27] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, "FiLM: Visual reasoning with a general conditioning layer," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018.